

Near-Field Integrated Sensing, Computing and Semantic Communication for Digital Twin-Assisted Vehicular Networks: A Reproduction

Reproducing Y. Yang et al., IEEE Transactions on Vehicular Technology, 2026

CS741 Next Generation Cloud Architecture

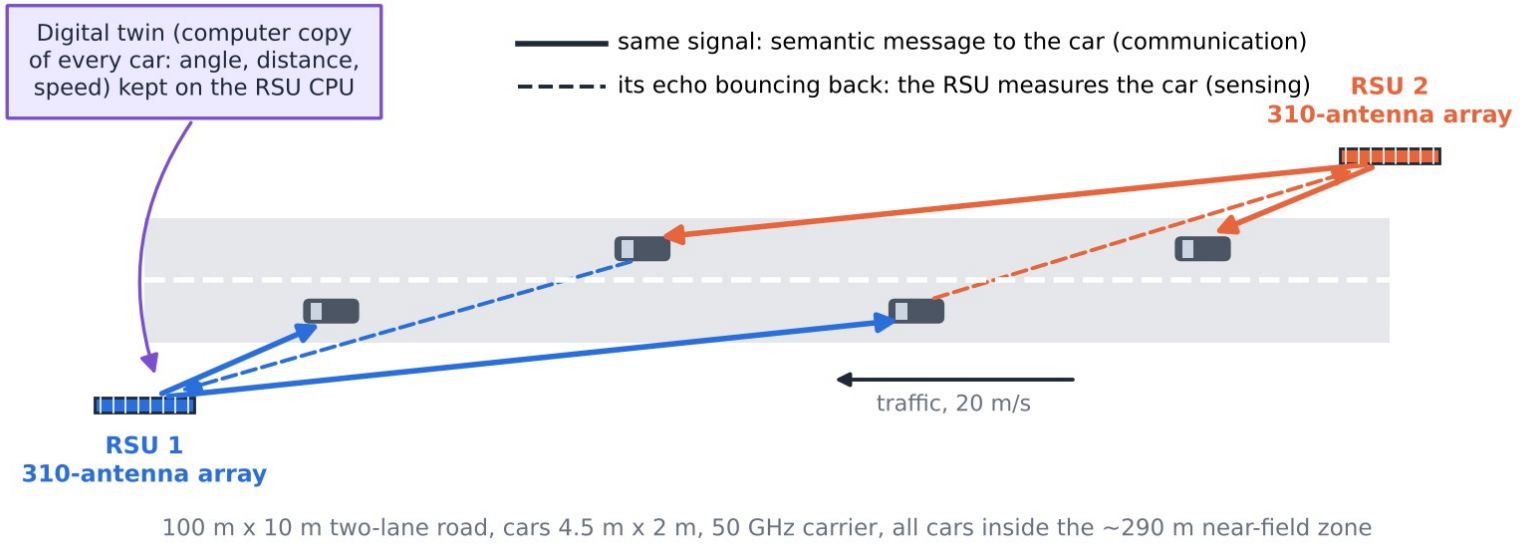
Yagnang Acharya (262IS037)
Akash Minz (262IS002)
Amul Gaurav (262IS005)

Mentor
Anshul Chauhan

Course Instructor
Dr. Sourav Kanti Addya

Department of Computer Science and Engineering
National Institute of Technology Karnataka, Surathkal | November 2026

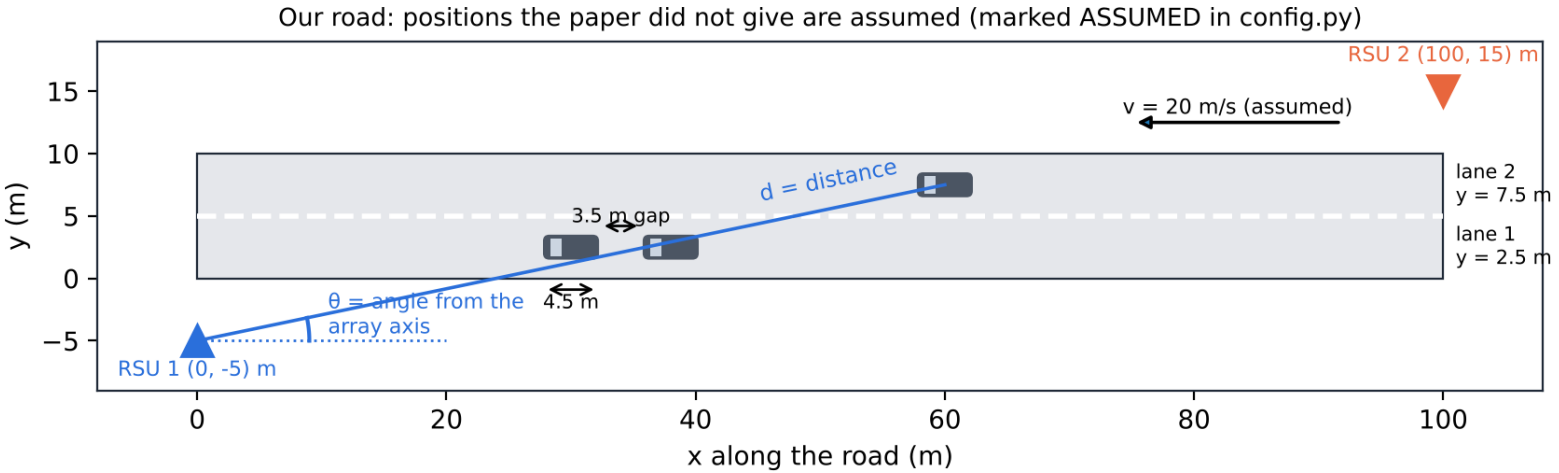
The question



100 m x 10 m two-lane road, cars 4.5 m x 2 m, 50 GHz carrier, all cars inside the ~290 m near-field zone

Paper's claim: about 20% more data rate than normal ISAC, with the same sensing accuracy

Our simulated road



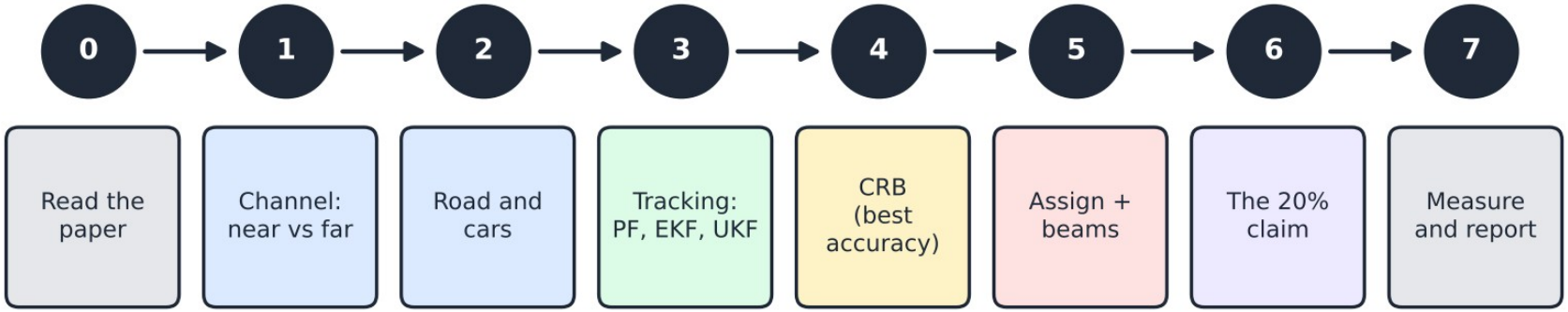
2 RSUs
roadside units

310
antennas each

50 GHz
radio frequency

5 to 20
cars on the road

How we worked

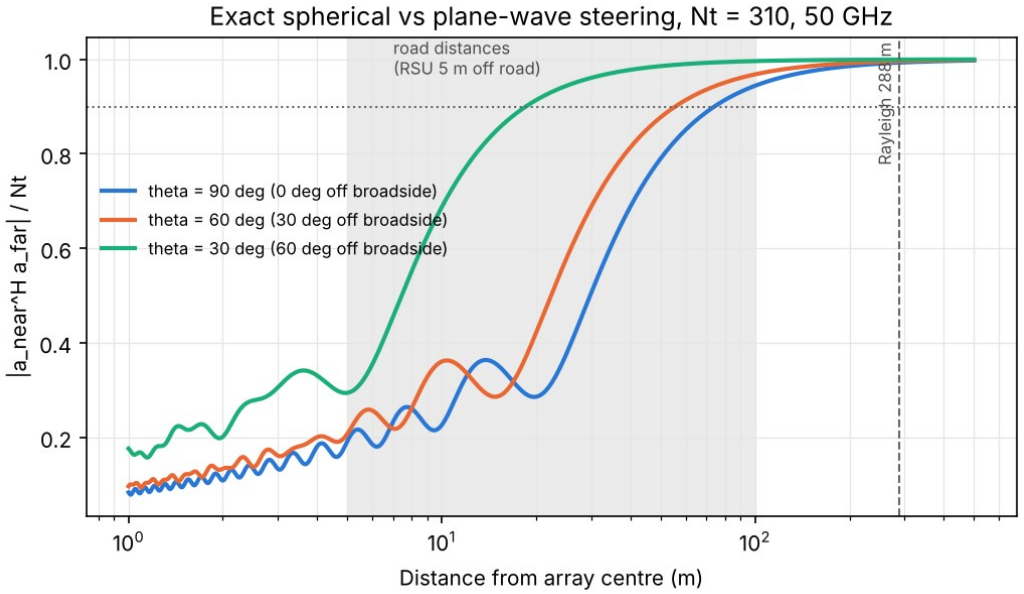


189 tests
all passing

5 seeds
per experiment

36
assumptions written down

Finding 1: the road is in the near field



288 m

near-field limit
(paper: about 300 m)

Reproduced
whole road is near field

Finding 2: tracking the cars

Three filters: particle filter (PF), EKF and UKF

Matches

Speed error
0.8 m/s
(paper: 0.8 m/s)

Differs

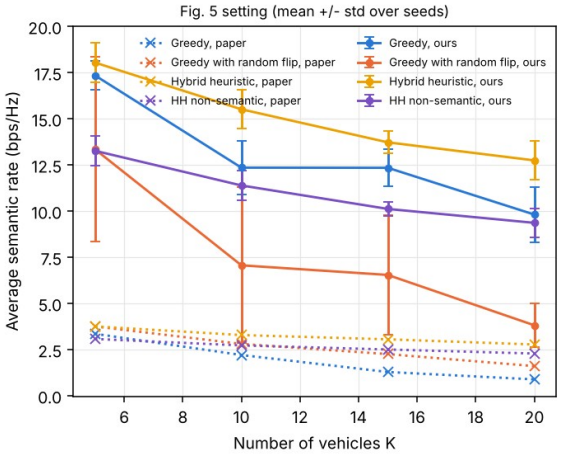
Angle error 1.0°
(paper: 1.6–2.1°)
Distance error 0.23 m
(paper: 0.08 m)

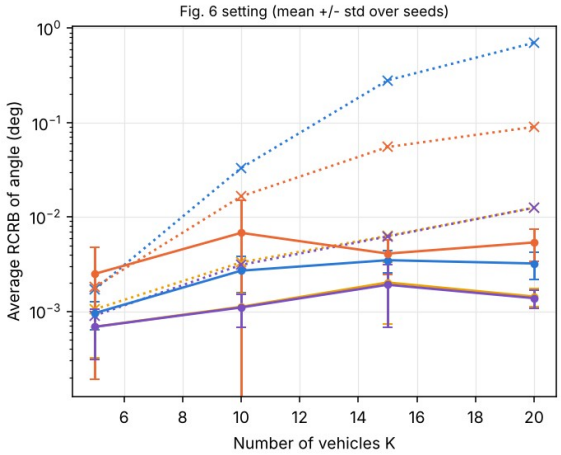
Differs

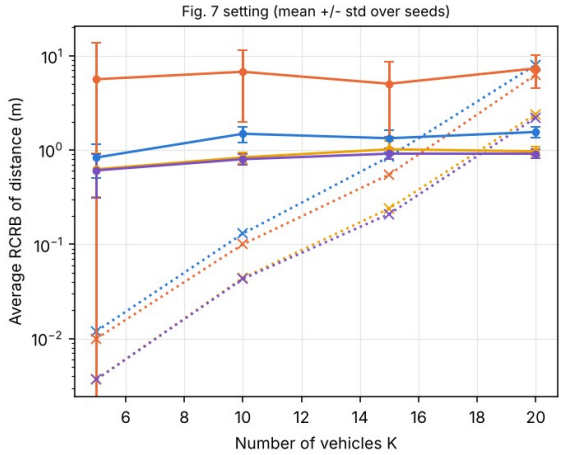
EKF and UKF are
as good as the PF
(paper: much worse)

Mean of 5 seeds, 5 to 20 cars

Finding 3: which RSU serves which car







Solid = ours, dotted = paper

Matches

Simulated annealing beats the nearest-RSU rule

Differs

Our data rates are 4–5 times the paper's

Finding 4: the 20% claim

36%, not 20%

Gain = 1.1 ÷ 0.81 − 1 = 36%: fixed by two numbers in the paper, not by the beamforming

1 Kb

Works for all cars
gain 36%

10 Kb

Works only for 5 cars

100 Kb

Impossible: one car
needs 6.7 GHz, RSU has 5.8

Overall: 16 claims checked

5	3	8
reproduced	partly reproduced	not reproduced

Two main reasons for the gaps

- The paper's path-loss formula has no wavelength term, so signals are too strong
- The semantic ratio always hits its lower limit (0.81), which fixes the gain at 36%

Conclusion

Holds: near field, speed tracking, simulated annealing beats nearest-RSU

Does not hold: the 20% gain; we get 36%, from a fixed multiplier

Limits: 10 Kb works only for 5 cars; 100 Kb is impossible

Next: physical path loss, missing units, cost comparison with Task 1

Thank you